上海科技大学
ShanghaiTech University

CS240 Algorithm Design and Analysis

Lecture 24

Randomized Algorithms
&
Approximation Algorithms

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Linear Programming

Linear Programming

- A farmer has the following problem.
  - He has 100 acres of land, on which he can plant wheat or barley or both.
  - He has 420 kg of fertilizer, and 160 kg of pesticide.
  - Each acre of barley requires 5 kg of fertilizer and 1 kg of pesticide.
  - Each acre of wheat requires 3 kg of fertilizer and 2 kg of pesticide.
  - Wheat sells for $3 per acre, barley sells for $4 per acre.
    - Actually, he could probably make $300 for wheat and $400 for barley. Choose $3 and $4 for simplicity.
- How many acres of wheat and barley should the farmer plant his field to maximize his income?
- Let w, b be acres of wheat and barley farmer plants.
- He wants to maximize 3w+4b, subject to the land, pesticide and fertilizer constraints.

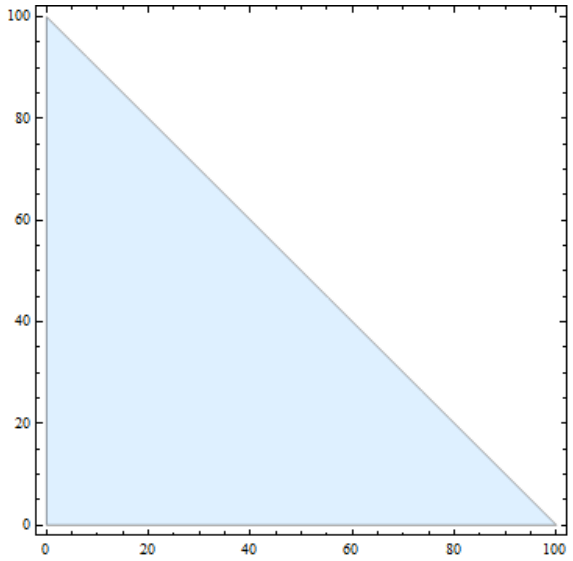
Linear Programming

- **maximize** 3w+4b **subject to**

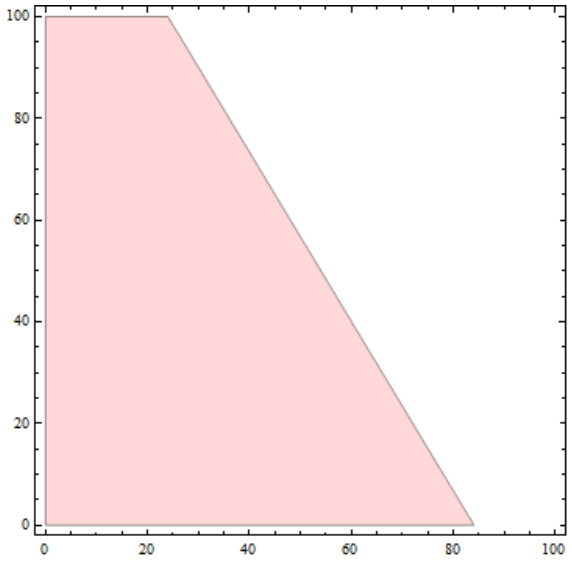
  $w+b \leq 100$ (land)

  $3w+5b \leq 420$ (fertilizer)

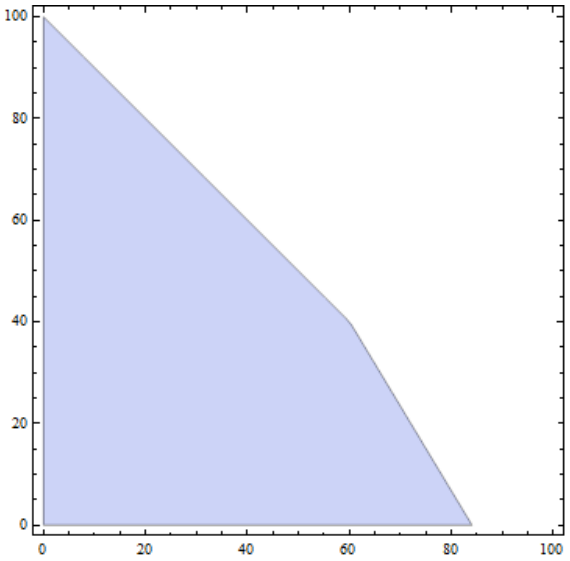
  $2w+b \leq 160$ (pesticide)



land constraint



fertilizer constraint



land + fertilizer constraints

Linear Programming

- **maximize** 3w+4b **subject to**

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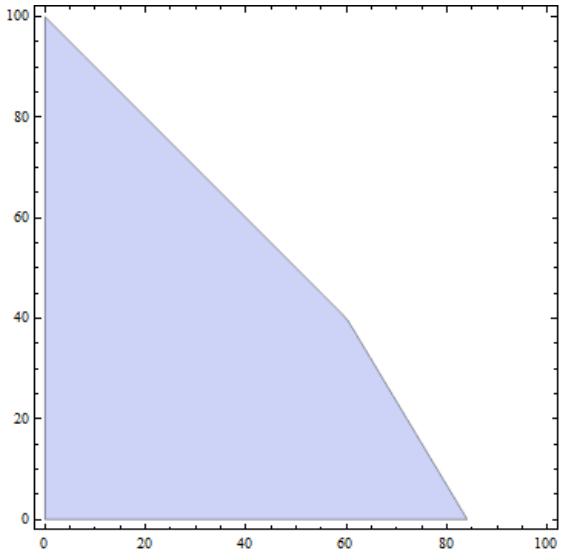
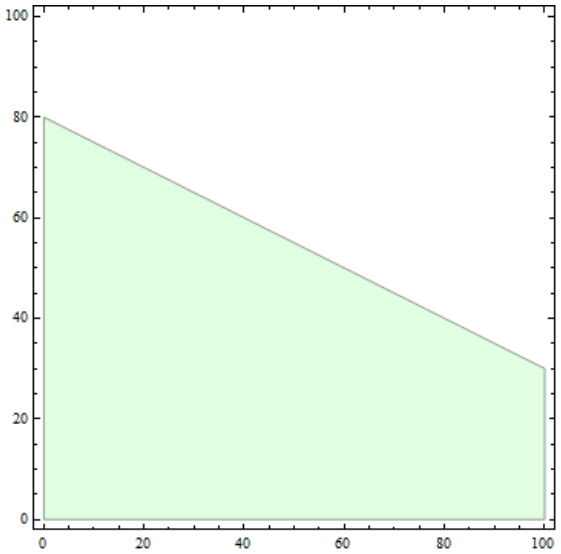
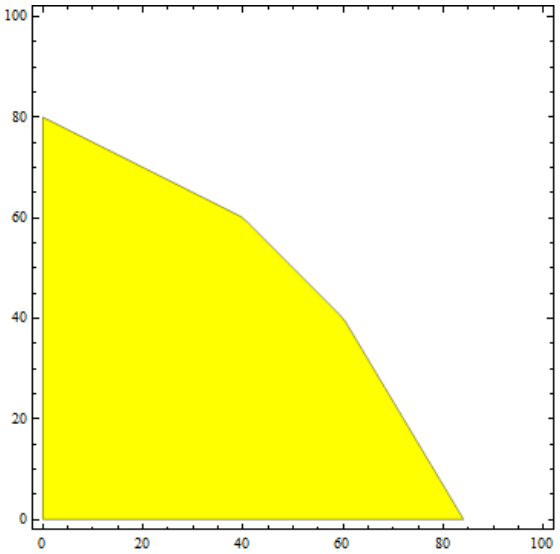


land + fertilizer constraints



pesticide constraints



all three constraints

Linear Programming

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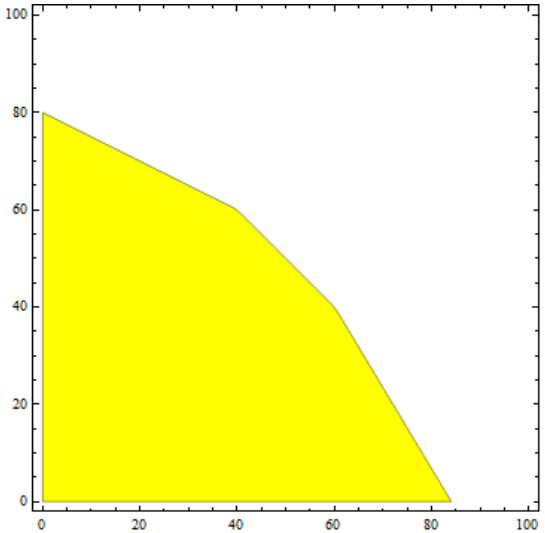
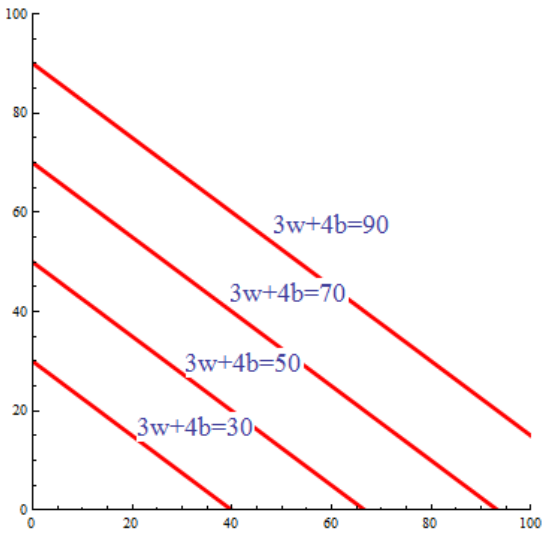
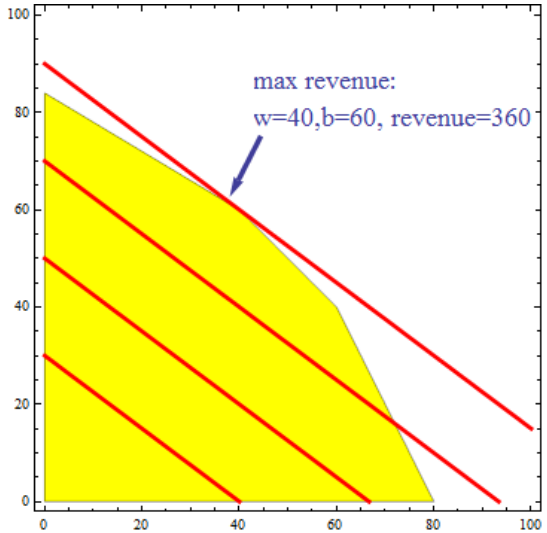
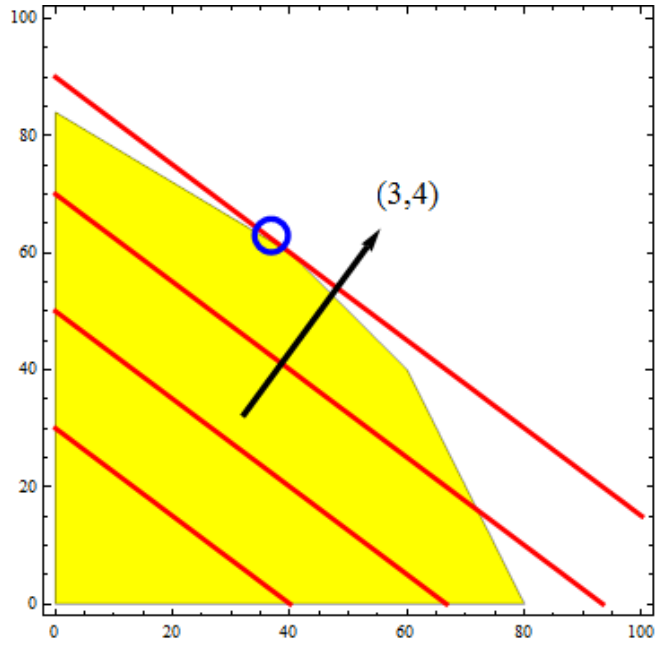






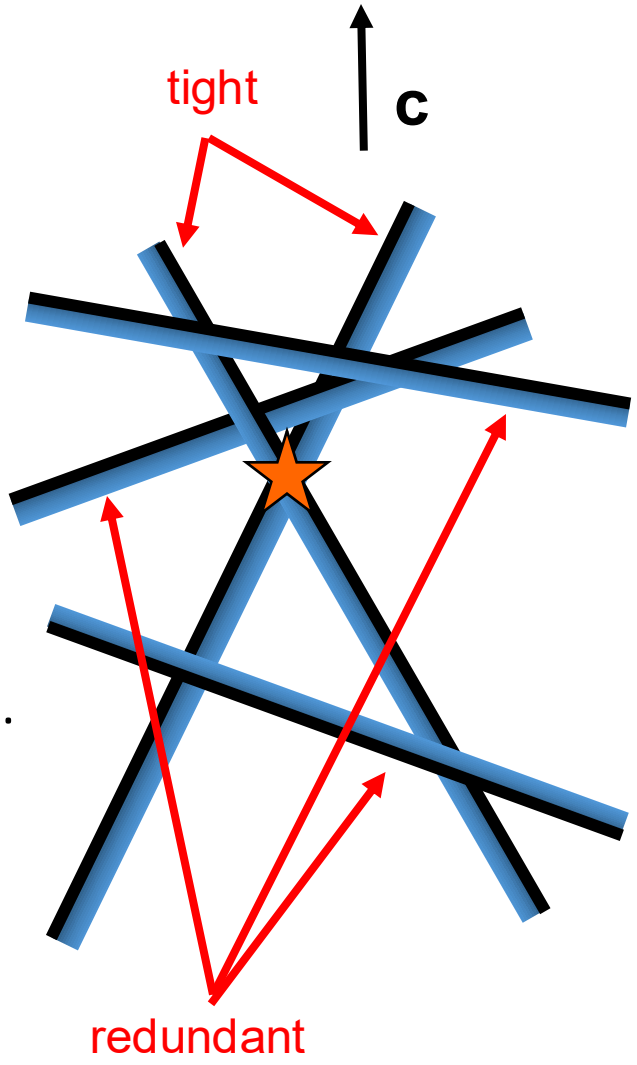
Linear Programming

- The feasible region is the area corresponding in which all the constraints are satisfied.
- Key Fact The optimum lies at an extreme point (corner).
- Find optimum by taking a line perpendicular to the direction pointed by the objective function, and shifting the line till when it will stop touching the feasible region.
- The optimum lies at the intersection of two constraints.
  - Call these the basis of the optimum.
  - For simplicity, assume constraints are general position, i.e., no 3 intersect at a point.



Randomized LP in 2D

- Since the optimum is defined by two constraints, the other constraints are redundant!
- A constraint is tight if the optimum lies on its defining line.
- Let H be set of n constraints. If pick random constraint, there's only 2/n probability it's tight.
- If constraint's not tight, we can discard it without changing optimum.
- How do we tell if it's tight?
  - For any constraint set G, let B(G) denote optimum.
- h∈H is redundant iff B(H)=B(H-{h}).
  - i.e., the optimum is the same with or without h. So opt doesn't lie on h.

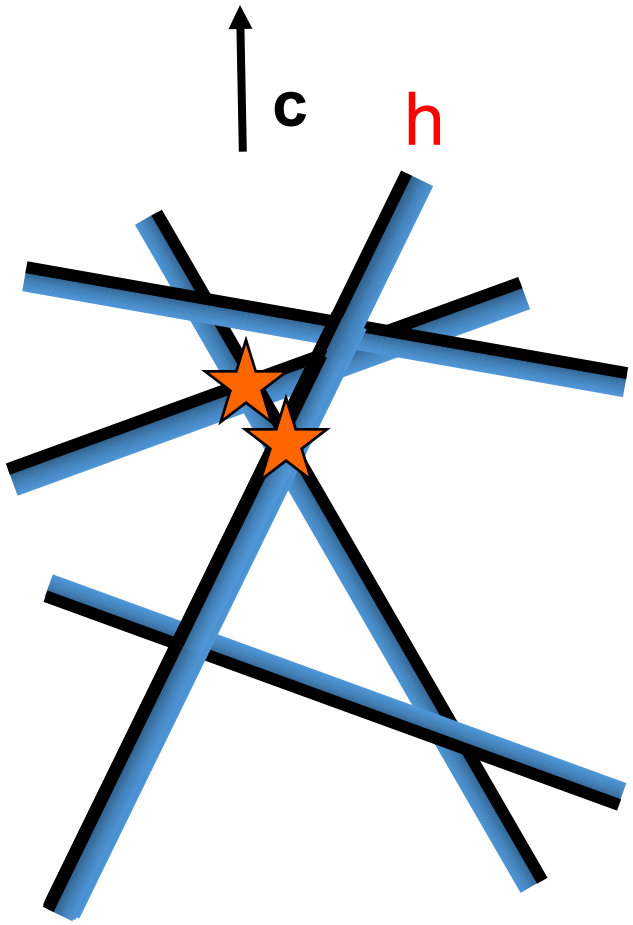


2D LP Algorithm

- If |H|=2, output intersection of the 2 halfplanes.
- Pick random constraint h ∈ H.
- Recursively find opt=B(H-{h}).
- If opt doesn't violate h, output opt.
  - opt violates h if opt lies outside h.
- Else project H-{h} onto h's boundary to obtain a 1D LP.
- Output the opt of the 1D LP.

Projection

- Given constraint h, let $\partial(h)$ be its boundary, i.e., the line defining h.
- Suppose B(H-{h}) violates h.
  - Then B(H) must lie on the boundary of h.
- Project a halfplane onto $\partial(h)$, reducing it to a line segment bounded on one or two sides.
- After projecting all H-{h} onto $\partial(h)$, we're left with a segment representing feasible region to 1D LP.
- Optimizing this is easy. The opt is one of the endpoints.



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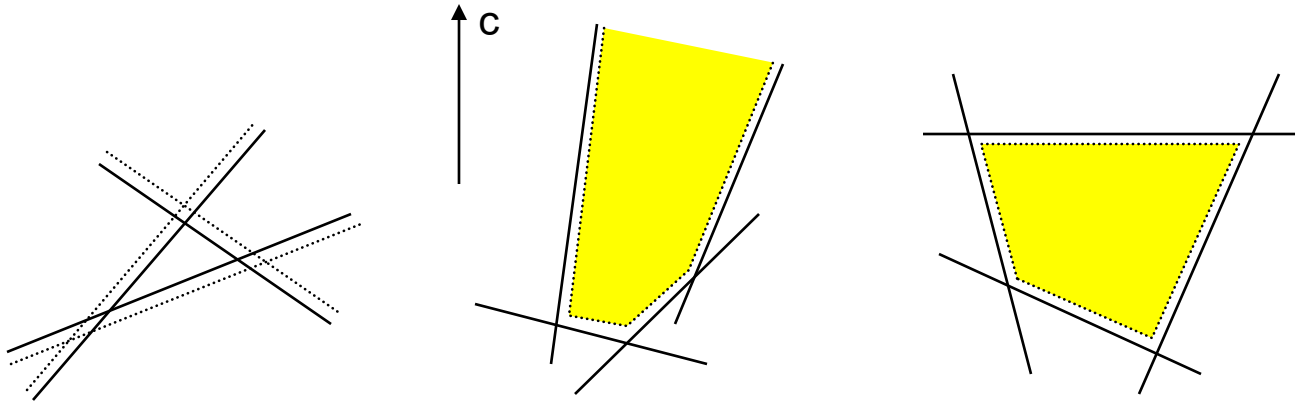
Analysis

- Let T(n) be expected time to solve 2D LP with n constraints.
- T(n) $\leq$ T(n-1)+O(1)+2/n(O(n)+O(1)).
- T(n-1) time recursively find opt=B(H-{h}).
- First O(1) is time to check whether opt violates h.
- There's 2/n probability opt violates h, in which case we project all constraints onto $\partial(h)$.
  - O(n) to project H-{h} onto $\partial(h)$.
  - Final O(1) to solve 1D LP.
- T(n) solves to O(n).
  - So we can solve 2D LP with n constraints in expected linear time.

Corner Cases

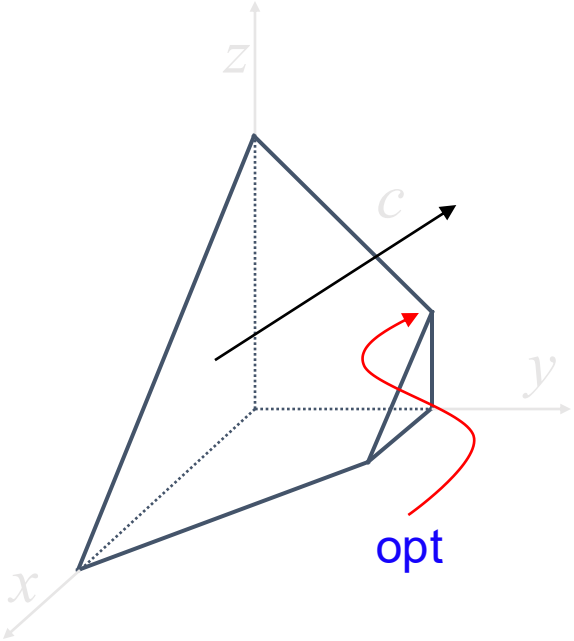
- For simplicity, we ignored several corner cases.
  - Infeasible means no points satisfy all the constraints.
    - Ex Constraints $x_1>1$ and $x_1<0$.
  - Unbounded means the optimum is infinite.
    - Ex Maximize x_2 s.t. $x_1+x_2>0$.
  - Non-unique optimum means an infinite number of points maximize the objective.
    - Ex Maximize x_2 s.t. $x_2 \leq 0$.
- Preprocess input to check for corner cases.

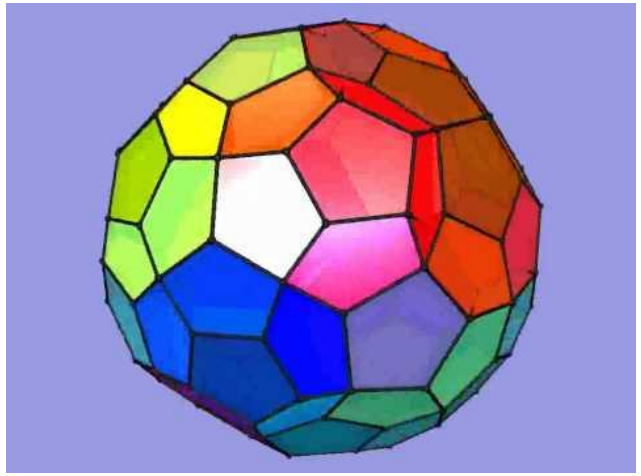


Infeasible Unbounded Non-unique optimum

Higher Dimensions

- In d>2 dimensions, lines become planes and each constraint corresponds to the space to one side of a plane, called a halfspace.
- The intersection of the halfspaces defines the feasible region.
  - This is a convex region called a polytope.
- Each extreme point (corner) of the polytope is the intersection of d halfspaces.
- The objective function defines a direction. Take a plane perpendicular to this direction and shift it till it stops touching feasible region.
- Hence optimum again lies at intersection of d halfspaces.
- Polytopes can be very complicated.





d-Dimensional LP Algorithm

- If |H|=d, output their intersection.
- Pick random constraint h∈H.
- Recursively find opt=B(H-{h}).
- If opt doesn't violate h, output opt.
- Else project H-{h} onto h's boundary to obtain a d-1 dimensional LP.
- Recursively solve the d-1 dim LP.

Analysis

- Let T(n,d) be expected time to solve d-dim LP with n constraints.
- T(n,d) $\leq$ T(n-1,d)+O(d)+d/n(O(dn)+T(n-1,d-1)).
- T(n-1,d) time recursively find opt=B(H-{h}).
- O(d) time to check whether opt violates h.
- There's d/n probability opt violates h.
  - Because opt is defined by d of the n halfspaces.
  - In this case we project all constraints onto $\partial(h)$.
  - O(dn) to project H-{h} onto $\partial(h)$.
  - We obtain a d-1 dim LP with n-1 constraints.
  - T(n-1,d-1) time to solve this.
- T(n,d) solves to O(d! n)
  - Linear in number of constraints.
  - Exponential in dimensions.

Matrix Formulation

- **maximize** 3w+4b **subject to**

  w+b $\leq$ 100

  3w+5b $\leq$ 420

  2w+b $\leq$ 160

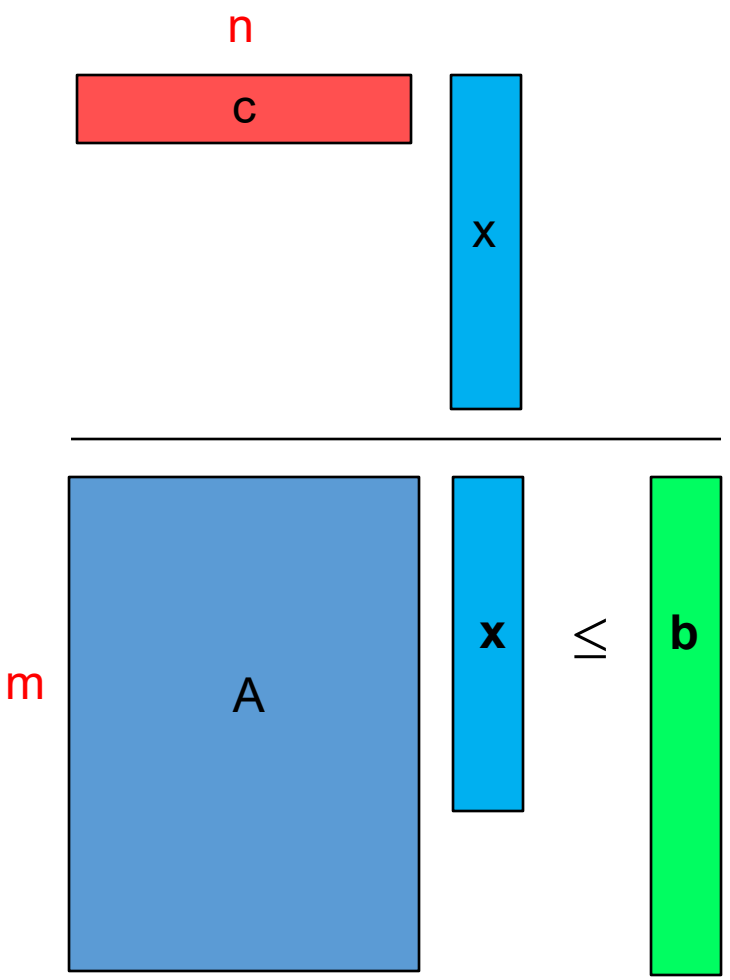
- $$\text{maximize} \quad [3,4]\cdot\begin{bmatrix} w \\ b \end{bmatrix} \text{ s.t.}$$

- $$\begin{bmatrix} 1 & 1 \\ 3 & 5 \\ 2 & 1 \end{bmatrix}\cdot\begin{bmatrix} w \\ b \end{bmatrix} \leq \begin{bmatrix} 100 \\ 420 \\ 160 \end{bmatrix}$$

  Let $x \in \Re^{n\times 1}, c \in \Re^{1\times n}, A \in \Re^{m\times n}, b \in \Re^{m\times 1}$.

  maximize $c \cdot x$ s.t.

  $A \cdot x \leq b$



Approximation Algorithms

Approximation Algorithms

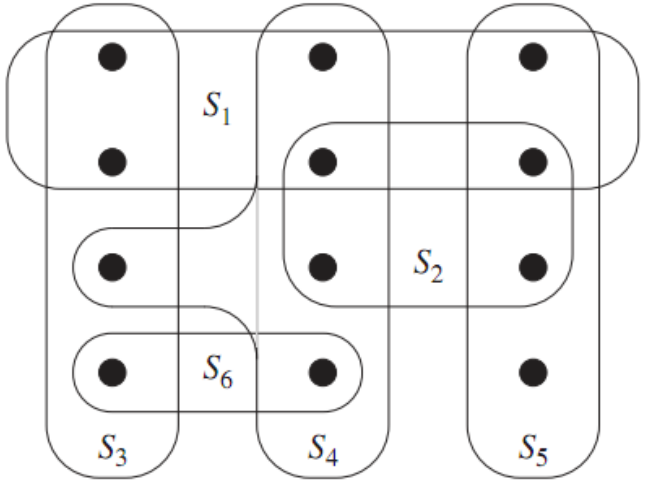
- Up to now, most of our algorithms have been exact, i.e., they find an optimal solution.
- But there are many problems for which we don't know how to find an optimal solution.
  - A key example is NP-complete problems. We don't know efficient algorithms for any NPC problem.
- Many such problems are important in practice. What do we do?
- If we can't get find the best answer, let's try for good enough.
- Approximation algorithms find an approximately optimal answer.

Approximation Ratio

- Let X be a maximization problem. Let A be an algorithm for X.
- Let a>1 be a constant.
- A is an a-approximation algorithm for X if A always returns an answer that's at least 1/a times the optimal.
  - Ex If X is max-flow, A is a 2-approx algorithm if it always returns a flow that's at least ½ the optimal.
  - The closer a is to 1, the better the approximation.
- If X is a minimization problem, A is an a-approximation algorithm for X if it always returns an answer that's at most a times larger than the optimal.
  - Ex If X is min-cut, A is a 2-approx algorithm if it always returns a cut that's at most 2 times the size of the optimal.
  - Again, the closer a is to 1, the better the approximation.

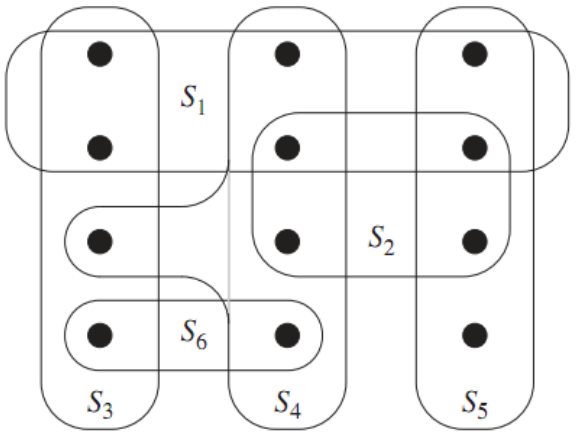
Coverings

- Suppose there's a set of teachers, and each can teach a certain set of classes.
  - Let S_i be the set of classes teacher i can teach.
- The entire set of classes is X.
- We want to pick the minimum set of teachers to teach all the classes.
  - Let T be set of teachers we pick.
  - We want $\cup_{i \in T} S_i = X$, and T to be the smallest possible.



Set Covering

- **Input** A collection F of sets. Each set has a cost. The union of all the sets is X.
- **Output** A subset G of F, whose union is X.
- **Goal** Minimize the total cost of the sets in G.



If all sets have same cost, S_3, S_4 and S_5 is a min cost set cover of X.

- Minimum cost set cover is NP-complete.
- We'll see a ln(n)-approximation algorithm, where n=|X|.

A Greedy Approximation Algorithm

- A natural greedy heuristic is to choose sets which cover points most cheaply.
  - For each set, let c be its cost, and m be the number of points it covers.
  - We want to use the set with the smallest c/m value, because this is the cheapest way to cover some new points.
- After we pick this set, remove all the points it covers. Then we consider the per unit cost of the remaining sets and again choose the cheapest.

A Greedy Approximation Algorithm

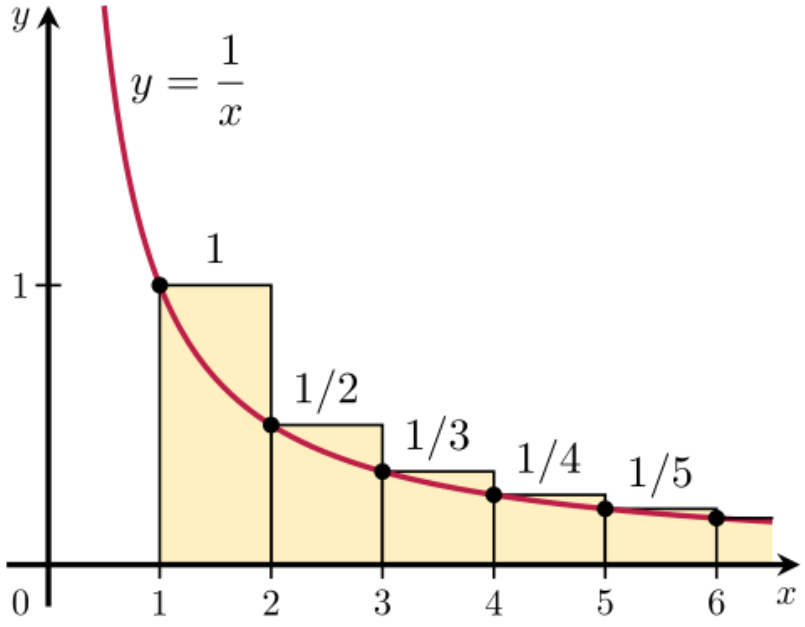
- F is the entire collection of sets. The union of these sets is X.
- Each set S in F has a cost cost(S).
- U is the set of elements of X we haven't covered yet.
- C is the set cover we eventually output.

- U = X
- C = ∅
- while $U \neq \emptyset$
  - choose $S \in F - C$ with min |cost(S)|/|S∩U|
  - C = C∪{S}
  - U = U – S
- output C

Per unit cost to cover new elements.

Proof of Correctness

- We always output a set cover, because the while loop continues till X is covered.
- We'll prove the approximation ratio is at most 1+1/2+1/3+…+1/n ≈ ln(n).
  - If the min cost of a set cover is V, our set cover costs at most ln(n)*V.
- The basic plan is to bound the cost of the set cover the algorithm outputs using the "average cost" per element.



Proof of Correctness

- Order the sets in C by when they're added to C, earliest set first.
  - Let the order be $S_1, S_2, \ldots, S_m$.
- Cost of the set cover is $L=\sum_i cost(S_i)$.
- Order the elements in X by when they're added, earliest element first.
  - Let the order be $e_1, e_2, \ldots, e_n$.
  - So, the first few e's are added by S_1, the next few added by S_2, etc.
  - Every element in X is in the list, because C covers X.

Proof of Correctness

- Let n_i be the number of new elements S_i covers.
  - So, n_i is the number of elements in S_i, but not in $S_1,\ldots,S_{i-1}$.
- Divide the cost of S_i evenly among the new elements it covers.
  - If e is newly covered by S_i, then cost(e)= cost(S_i)/n_i.

$$\sum_{\mathrm{k}} cost(e_k) = \sum_i n_i * \frac{cost(S_i)}{n_i} = \sum_i cost(S_i) = L$$

  - Every element is covered by some S_i, and S_i covers n_i new elements.
- **We'll prove cost(e_k) $\leq$ OPT/(n-k+1), for any k.**
- Suppose this is true, then

$$L = \sum_k cost(e_k) \leq \sum_k OPT/(n-k+1) \approx \ln(n) * OPT$$

The Per Element Cost

- Let's focus on some element e_k, and let S_j be the set which covers e_k for the first time.
- Let $C_1,\ldots,C_m$ be the sets in an optimal cover, each of which covers some elements of U= $\{e_k,e_{k+1},e_{k+2},\ldots,e_n\}$.
  - Let $n'_1,\ldots,n'_m$ be the number of elements of U which $C_1,\ldots,C_m$ cover.
- **Obs 1: $\sum_i n'_i \geq$ n-k+1.**
  - Because $C_1,\ldots,C_m$ cover U.
- **Obs 2: $\sum_i cost(C_i) \leq$ OPT.**
  - Because $C_1,\ldots,C_m$ are a subset of an optimal cover, which has cost OPT.

The Per Element Cost

- **Obs 3** None of $C_1,\ldots,C_m$ are among $S_1,\ldots,S_{j-1}$.
  - If some C_i is among $S_1,\ldots,S_{j-1}$, then since C_i covers some e in U, e would be covered by $\{S_1,\ldots,S_{j-1}\}$. So, e would be among the first k-1 elements covered. Contradiction.
- **Obs 4** There exists some C_i among $C_1,\ldots,C_m$ with $\frac{cost(C_i)}{n'_i} \leq OPT/(n-k+1)$.
  - If every C_i in $C_1,\ldots,C_m$ has $\frac{cost(C_i)}{n'_i} \geq OPT/(n-k+1)$, then

$$OPT \geq \sum_i cost(C_i) = \sum_i n'_i * \frac{cost(C_i)}{n'_i} > \sum_i n'_i * \frac{OPT}{n-k+1} \geq OPT/(n-k+1) \sum_i n'_i \geq \frac{OPT}{n-k+1} * (n-k+1) = OPT$$

Contradiction.

Proof of Approximation Ratio

- **Lemma** $\boldsymbol{cost(S_j)/n_j \le OPT/(n-k+1)}$
- **Proof** When choosing S_j, the only sets the algorithm is not allowed to choose are $S_1,\ldots,S_{j-1}$.
  - By obs 3, $C_1,\ldots,C_m$ aren't in $S_1,\ldots,S_{j-1}$.
  - By obs 4, there's some C_i in $C_1,\ldots,C_m$, with $\frac{cost(C_j)}{n'_i} \le OPT/(n-k+1)$.
  - S_j was chosen so that $cost(S_j)/n_j$ is min among all sets not in $S_1,\ldots,S_{j-1}$.
  - So $\frac{cost(S_j)}{n_j} \le \frac{cost(C_i)}{n'_i} \le OPT/(n-k+1)$.
- Since $\frac{cost(S_j)}{n_j} = cost(e_k)$, we have $cost(e_k) \le OPT/(n-k+1)$.
- The approximation ratio follows because

$$\boldsymbol{L = \sum_k cost(e_k) = \sum_k \frac{OPT}{n-k+1} \approx In(n) * OPT}$$